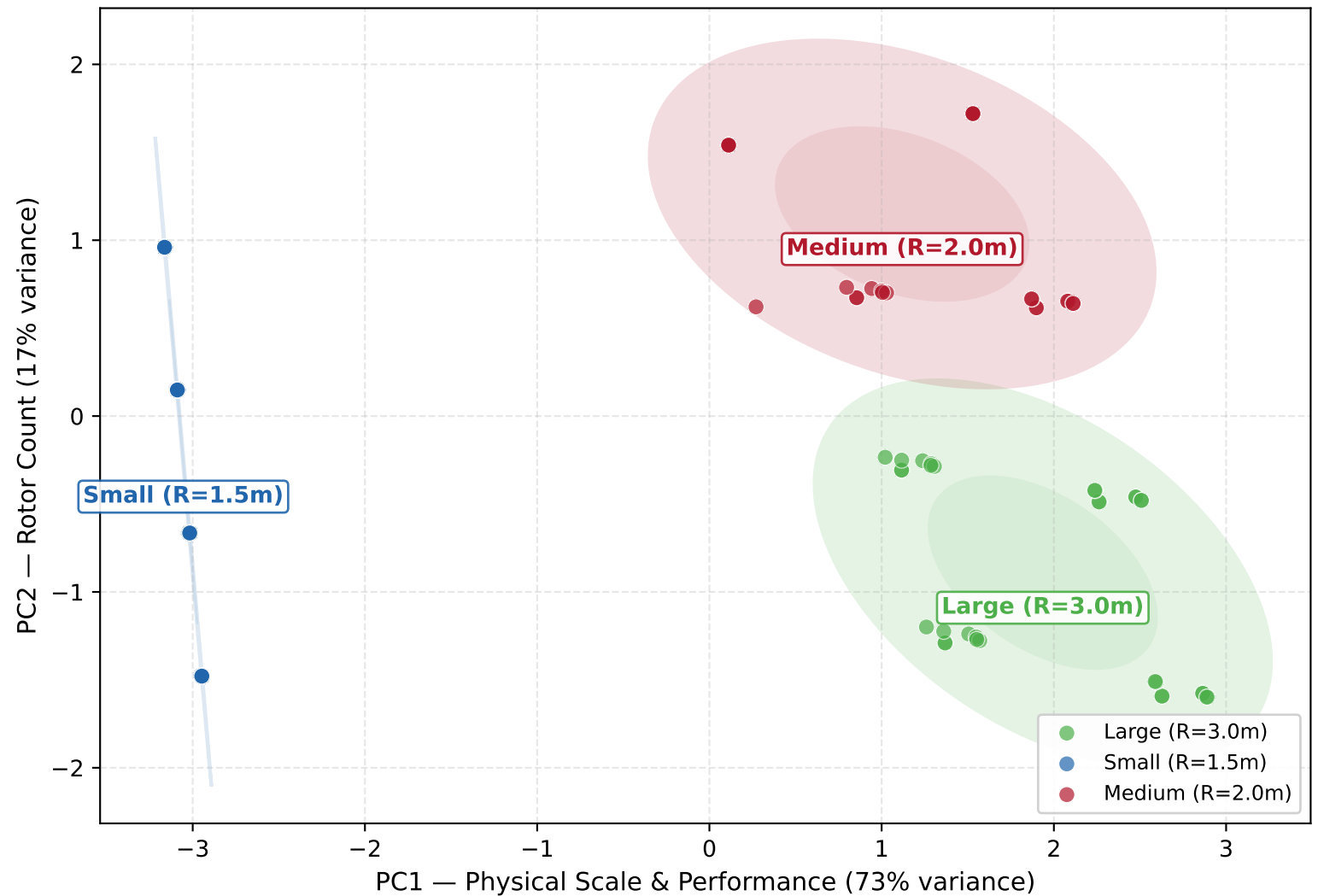


K-Means Clusters in PC Space: Natural Design Families



IMPLICATIONS: K-means (k=3) identifies three natural design families separated primarily by rotor radius — small (R=1.5m), medium (R=2.0m), large (R=3.0m). Within each cluster, configurations vary by rotor count (PC2) and elevation but remain clearly grouped. The clusters are well-separated with minimal overlap — radius is a first-order design discriminator. Manufacturing implication: build 3 rotor sizes, not 96 configurations. Within each size class, offer 1-4 rotor stacking options for different tension requirements. Data: 96 configurations from BEM v2.0 sweep.